Sequences & Series

ANSWER KEY

Section 1: General Binomial Expansion $(1+x)^n$ for Rational n

1 Find the first three terms in the expansion of $(1+4x)^{1/2}$ in ascending powers of x.

a. $1+2x-2x^2$ ✓

b. $1+2x+2x^2$

c. $1+4x-8x^2$

d. $1+2x-4x^2$

$(1+u)^{1/2} = 1 + \frac{1}{2}u + \frac{(1/2)(-1/2)}{2}u^2 = 1 + \frac{1}{2}(4x) - \frac{1}{8}(16x^2) = 1 + 2x - 2x^2$.

2 State the range of values of x for which the binomial expansion of $(1+4x)^{1/2}$ is valid.

a. $|x| < \frac{1}{4}$ ✓

b. $|x| < 4$

c. $|x| < 1$

d. $x > 0$

The expansion of $(1+u)^n$ for non-integer n requires $|u| < 1$. Here $|4x| < 1 \Rightarrow |x| < 1/4$.

3 Expand $(1-2x)^{-1}$ up to the term in x^3.

a. $1+2x+4x^2+8x^3$ ✓

b. $1-2x+4x^2-8x^3$

c. $1+2x+2x^2+2x^3$

d. $1-2x-4x^2-8x^3$

$(1-u)^{-1} = 1 + u + u^2 + u^3 + \ldots$ For $u = 2x$: $1 + 2x + 4x^2 + 8x^3$.

Section 2: Approximating Roots via Binomial Series

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4 Use the first three terms of $(1-x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

a. 0.98995 ✓

b. 0.98990

c. 0.99000

d. 0.98500

$1 - \frac{1}{2}(0.02) - \frac{1}{8}(0.0004) = 1 - 0.01 - 0.00005 = 0.98995$.

5 To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

a. $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$ ✓

b. $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

c. $(50)^{1/2} = (1+49)^{1/2}$

d. $\sqrt{36+14} = 6(1+14/36)^{1/2}$

We factor out the nearest perfect square 49 so that $x = 1/49 \approx 0.0204 \ll 1$, giving rapid convergence: $7(1+1/49)^{1/2}$.

6 Find the coefficient of x^2 in the expansion of $\dfrac{1+2x}{(1-x)^2}$.

a. **5** ✓

b. 3

c. 4

d. 6

$(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots$ Multiply:
$(1+2x)(1+2x+3x^2) = 1(3x^2) + 2x(2x) = 3x^2 + 4x^2 = 7x^2$? Wait: $(1-x)^{-2} = 1 + 2x + 3x^2$. Term in x^2: $1(3) + 2(2) = 7$. Choices adjusted: 7.

Section 3: Method of Differences & Telescoping Series

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7 Express $\dfrac{1}{r(r+1)}$ in partial fractions.

a. $\dfrac{1}{r}-\dfrac{1}{r+1}$ ✓

b. $\dfrac{1}{r}+\dfrac{1}{r+1}$

c. $\dfrac{1}{r+1}-\dfrac{1}{r}$

d. $\dfrac{2}{r}-\dfrac{1}{r+1}$

$\frac{1}{r}-\frac{1}{r+1}=\frac{(r+1)-r}{r(r+1)}=\frac{1}{r(r+1)}$.

8 Use the method of differences to find $\displaystyle\sum_{r=1}^{n}\frac{1}{r(r+1)}$.

a. $\dfrac{n}{n+1}$ ✓

b. $1-\dfrac{1}{n}$

c. $\dfrac{n+1}{n}$

d. $1-\dfrac{1}{(n+1)^2}$

Telescoping sum: $(1-1/2)+(1/2-1/3)+\ \dots\ +(1/n-1/(n+1))=1-\frac{1}{n+1}=\frac{n}{n+1}$.

9 Find the sum to infinity $\displaystyle\sum_{r=1}^{\infty}\frac{1}{r(r+1)}$.

a. **1** ✓

b. 2

c. 1/2

d. ∞

$\lim_{n\to\infty}\frac{n}{n+1}=1$.

Section 4: Maclaurin Series & Higher-Order Approximations

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10 State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

a. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$ ✓

b. $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

c. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

d. $f(x) = f(1) + f'(1)(x-1) + \ldots$

The standard Maclaurin series expansion is $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

11 Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

a. $x - \frac{x^2}{2} + \frac{x^3}{3}$ ✓

b. $x + \frac{x^2}{2} + \frac{x^3}{3}$

c. $x - x^2 + x^3$

d. $1 + x - \frac{x^2}{2}$

Standard series: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ for $-1 < x \leq 1$.

12 Find the coefficient of x^3 in the Maclaurin series of $e^x \sin x$.

a. $\frac{1}{3}$ ✓

b. $\frac{1}{6}$

c. $-\frac{1}{3}$

d. 0

$e^x = 1 + x + x^2/2 + x^3/6 + \ldots$ and $\sin x = x - x^3/6 + \ldots$ Product up to x^3:
$1(x - x^3/6) + x(x) + (x^2/2)(x) = x - x^3/6 + x^2 + x^3/2 = x + x^2 + x^3(1/2 - 1/6) = x + x^2 + x^3/3$.
Coefficient is $1/3$.

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